Blueprint

The order generated by a column of the $SU(n)_k$ S-matrix is locally a numerical semigroup ring (Part I)

Algebraic number theory · orders generated by character values

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At a glance

What the paper contributes

A column μ of the modular S-matrix of $SU(n)_k$, normalised by its first entry, generates a ring A_μ of character values. Its field was known (Gannon–Schopieray); the paper determines the **ring** when the height $p = n + k$ is prime.

- **The decomposition** (Lemma 1, Proposition 2): $A_\mu = \mathbb{Z}[\zeta_m] \otimes C_T$, a maximal cyclotomic factor fixed by the charge under the centre, and an order C_T that depends only on the reduced spectrum $T \subset \mathbb{F}_p$; the index is a power of p.
- **The exact semigroup** (Theorem 3, Corollary 5): locally at p, C_T is a monomial ring whose valuations form a numerical semigroup S_T, written by Gauss sums and Stickelberger's theorem; the index is $p^{\varphi(m)\delta}$, δ the number of gaps.
- **Read in $\mathbb{F}_p$** (Theorem 4): when $p > (n-1)(n-2)$, and for every p when $n \le 12$, S_T is generated by e and the j with $M_{hj}(T) \neq 0$.
- **One semigroup, every property** (Corollaries 6–8, Propositions 9–10): Gorenstein ⇔ S_T symmetric (Kunz), exponent p when the conductor is small, the vacuum is maximal, level–rank keeps S_T, the conformal weight sees M_2.
- **Counts and extremes** (Propositions 11–12, §6): a closed count of the maximal columns, the largest defect $(n-1)(n-2)/2$ attained exactly by the trinomials $X^n + aX + b$, and the first non-Gorenstein column at prime height, $SU(5)_{12}$.

🚀 **Start here: Proposition 2**: the ring of a column is Z[ζ_m] ⊗ C_T, and its index is a power of p.

Main results:

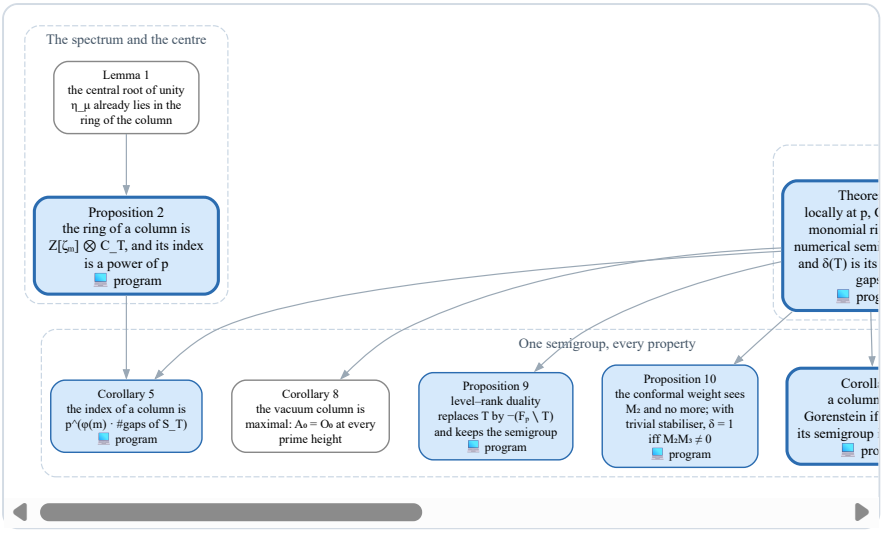
- **Proposition 2**: the ring of a column is Z[ζ_m] ⊗ C_T, and its index is a power of p
- **Theorem 3**: locally at p, C_T is the monomial ring of the numerical semigroup S_T, and δ(T) is its number of gaps
- **Theorem 4**: when $p > (n-1)(n-2)$, or $n \le 12$, the semigroup is read in $\mathbb{F}_p$ from the nonzero power sums $Mh_j(T)$
- **Corollary 6**: a column ring is Gorenstein if and only if its semigroup is symmetric

12 results | ✅ **0** formalized in Lean | ◐ **1** with a part in Lean

💻 **10** checked by a program | written proof only | (a result can carry several tags)

🗺️ Map of the results (click a box to go to it)

Dependencies in the written proof: an arrow A → B means that this proof of B uses A. Every result rests on its written proof; the tags record what was checked besides, and each statement's “Evidence & scope” block says how far. The colour of a box is its strongest tag, in the order above; the tags in the box give the rest. Boxes with a thick border are the main results; click a box to read the statement.



📐 Setting (copied from the paper, §1–§3)

(Abstract) Fix a column μ of the modular S-matrix of $\mathrm{SU}(n)_k$ and let A_μ be the ring generated by the ratios $S_{\lambda\mu}/S_{0\mu}$: the character values of $\mathrm{SU}(n)$ at the element of finite order that labels μ. *(§1)* For $\mathrm{SU}(n)$ at level k and integrable weights λ, μ,

$$\frac{S_{\lambda\mu}}{S_{0\mu}} = \mathrm{ch}_\lambda(g_\mu), \qquad g_\mu = \exp\!\Big(2\pi i\,\frac{\mu+\rho}{k+n}\Big).$$

Throughout, p is an odd prime, $n \geq 2$, $k = p - n \geq 1$, $\zeta_N = e^{2\pi i/N}$, and for a column μ we write $\mathcal{O}_\mu$ for the integers of the field K_μ of A_μ. *(§2)* Write $\ell_i = \sum_{j\geq i}\mu_j + n - i$ $(1 \leq i \leq n)$, $L = \sum_i \ell_i$, and fix integers u, v with $up + vn = 1$. The eigenvalues of g_μ on $\mathbb{C}^n$ are $\zeta_{np}^{n\ell_i - L}$, and

$$\zeta_{np}^{\,n\ell_i - L} = \eta_\mu\, \zeta_p^{\,t_i}, \qquad \eta_\mu = \zeta_n^{-uL}, \qquad t_i = \ell_i - vL \bmod p.$$

The ℓ_i are distinct in $[0, p-1]$, so $T = T_\mu = \{t_1, \dots, t_n\} \subset \mathbb{F}_p$ has n elements, and $\sum_i t_i = L(1 - nv) = Lup \equiv 0$: the spectrum is *centred*. The scalar η_μ is a root of unity of order $m = m_\mu$ dividing n. *(proof of Lemma 1)* With $y_t = \zeta_p^t$, …Let $C_T = \mathbb{Z}[e_1(y_T), \dots, e_{n-1}(y_T)] \subset \mathbb{Z}[\zeta_p]$, $H_T = \{a \in \mathbb{F}_p^\times : aT = T\}$ of order h, and $E_T = \mathbb{Q}(\zeta_p)^{H_T}$, of degree $e = (p-1)/h$. From now on $\delta(T) = v_p[\mathcal{O}_{E_T} : C_T]$. *(§3)* Let $\omega : \mathbb{F}_p^\times \to \mathbb{Z}_p^\times$ be the Teichmüller character, and put

$$B_j(T) = \sum_{t\in T,\, t\neq 0} \omega(t)^{hj} \in \mathbb{Z}_p, \qquad b_j = v_p(B_j) \in \{0, 1, \dots, \infty\} \qquad (1 \leq j < e).$$

Since $\omega(t) \equiv t$, $b_j = 0$ exactly when the power sum $M_{hj}(T) = \sum_{t\in T} t^{hj}$ is nonzero in $\mathbb{F}_p$.

2 The spectrum and the centre

Lemma 1. (the centre is in the ring) p. 3

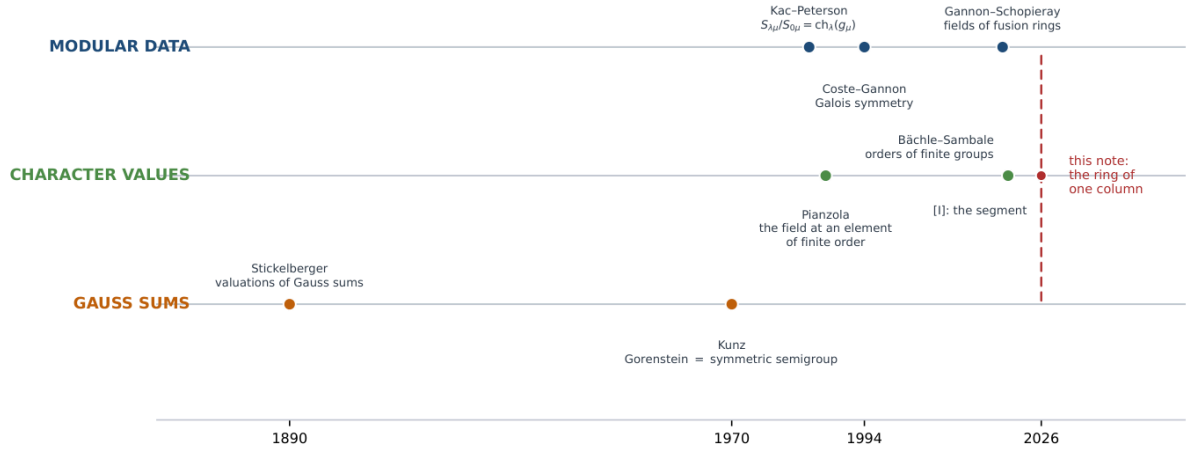
> $\eta_\mu \in A_\mu$.

↳ Used in: Proposition 2 (p. 4)

Proof

The complete symmetric function h_{p-n} is the character of $\mathrm{Sym}^{p-n}\,\mathbb{C}^n$. With $y_t = \zeta_p^t$, $\prod_{t\in T}(1-y_tX)\prod_{t\notin T}(1-y_tX) = 1-X^p$, so $\sum_j h_j(y_T)X^j = \prod_{t\notin T}(1-y_tX)/(1-X^p)$ and $h_{p-n}(y_T)$ is the top coefficient of the numerator, $(-1)^{p-n}\prod_{t\notin T} y_t = (-1)^{n-1}$, because $\sum_{t\in\mathbb{F}_p} t = 0 = \sum_{t\in T} t$. By homogeneity $\mathrm{ch}_{h_{p-n}}(g_\mu) = (-1)^{n-1}\eta_\mu^{p-n} = \pm\eta_\mu^p$, and p is invertible modulo the order of η_μ.

Figure 1



Three lanes, and this note at the crossing. *Modular data*: from Kac–Peterson's formula to the fields of Gannon and Schopieray; the question ends at a field. *Character values*: the field at an element of finite order (Pianzola), the order for finite groups (Bächle–Sambale), the order at the principal element ([10]). *Gauss sums and semigroups*: Stickelberger's valuations make the ring monomial, and Kunz reads the Gorenstein property off its semigroup.

Proposition 2. (the decomposition) p. 3

$A_\mu = \mathbb{Z}[\zeta_m]\otimes C_T$, $K_\mu = \mathbb{Q}(\zeta_m)E_T$, *and*

$$\setlength\fboxsep6pt\setlength\fboxrule0.6pt\ [\mathcal{O}_\mu : A_\mu] = [\mathcal{O}_{E_T} : C_T]^{\varphi(m)},$$

a power of p.

↳ Used in: Corollary 5 (p. 7)

Proof

The fundamental characters generate the representation ring, so $A_\mu = \mathbb{Z}[e_j(g_\mu)]$ with $e_j(g_\mu) = \eta_\mu^j e_j(y_T)$; with Lemma 1, $A_\mu = \mathbb{Z}[\eta_\mu]\cdot C_T$. As $\gcd(m,p) = 1$, $\mathbb{Q}(\zeta_m)$ and E_T are linearly disjoint with coprime discriminants, so $\mathcal{O}_{K_\mu} = \mathbb{Z}[\zeta_m]\otimes\mathcal{O}_{E_T}$ and $\mathbb{Z}[\zeta_m]\cdot C_T \cong \mathbb{Z}[\zeta_m]\otimes C_T$; the index is $[\mathcal{O}_{E_T} : C_T]^{\varphi(m)}$. It remains to see that C_T is maximal at every prime $r \neq p$. The prime r is unramified in E_T, so $\mathcal{O}_{E_T}/r$ is a finite étale $\mathbb{F}_r$-algebra, whose geometric points are the e embeddings σ_a, $a \in \mathbb{F}_p^\times/H_T$, followed by $\zeta_p \mapsto \xi$, a primitive p-th root of unity in $\overline{\mathbb{F}}_r$. The image of C_T separates

them: $e_j(y_T)$ goes to the coefficients of $\prod_{t\in T}(X-\xi^{at})$, and $\{\xi^{at}\}=\{\xi^{a't}\}$ forces $aT=a'T$. A subalgebra of a finite étale algebra that separates its geometric points is everything, so $C_T+r\mathcal{O}_{E_T}=\mathcal{O}_{E_T}$, and Nakayama gives $C_T\otimes\mathbb{Z}_r=\mathcal{O}_{E_T}\otimes\mathbb{Z}_r$.

Evidence & scope

- **Program:** the index [O_μ : A_μ] = p^φ(m)δ as a lattice index in all 662 columns of SU(3)_4, SU(4)_3, SU(3)_8, SU(8)_3, SU(5)_6, SU(6)_5. `nivel_rango.py`

3 The semigroup of a spectrum

Theorem 3. (the exact semigroup) p. 4

Let $R=C_T\otimes\mathbb{Z}_p$ and let $\theta=g_1^{\,h}$, with g_1 the Gauss sum below, a uniformizer of $E_T\otimes\mathbb{Q}_p$ on which the Galois group acts by a character. Then R is the monomial ring

$$R=\sum_{s\in S_T}\mathbb{Z}_p\,\theta^s,\qquad \setlength\fboxsep6pt\setlength\fboxrule0.6pt\ S_T=\big\langle\, e,\ \ j+$$

the valuations of its nonzero elements form the numerical semigroup S_T, and $\delta(T)=\#(\mathbb{N}_0\setminus S_T)$, the number of gaps.

Classical and new: built from Stickelberger's valuations of Gauss sums [Sti90; Rob01, Cor. 1] and Selmer's formula for the genus [Sel77].

↳ Used in: Corollary 5 (p. 7), Corollary 6 (p. 7), Corollary 8 (p. 8), Proposition 9 (p. 8), Proposition 10 (p. 9), Proposition 11 (p. 9), Proposition 12 (p. 10)

Proof

Power sums. By Newton's identities e_j is a polynomial in $p_1,\dots,p_j$ with denominators dividing $j!$, invertible in $\mathbb{Z}_p$ as $j\le n<p$; and every power sum is an integral polynomial in the e_j. Hence $R=\mathbb{Z}_p[p_r(y_T):r\ge1]=\mathbb{Z}_p[\sigma_a(\alpha):a\in\mathbb{F}_p^\times]$ with $\alpha=\sum_{t\in T}\zeta_p^t$.

Eigenvectors. R is stable under $\mathrm{Gal}(\mathbb{Q}_p(\zeta_p)/\mathbb{Q}_p)\cong\mathbb{F}_p^\times$, and $p-1$ is a unit, so the projectors $P_r=\frac{1}{p-1}\sum_a\omega(a)^{-r}\sigma_a$ act on R, and R is generated as a $\mathbb{Z}_p$-algebra by the $P_r\alpha$. Substituting $b=at$,

$$P_r\alpha=\frac{1}{p-1}\,B'_r\,g_r,\qquad g_r=\sum_{b\in\mathbb{F}_p^\times}\omega(b)^{-r}\zeta_p^b,\qquad B'_r=\sum_{t\in T,\,t\neq0}\omega(t)^r,$$

and $B'_r=0$ unless $h\mid r$, because T is H_T-stable. By Stickelberger's theorem $v_\pi(g_r)=r$ for $0\le r<p-1$, with $\pi=\zeta_p-1$ [15 , Cor. 1, p. 168] .

The uniformizer. Put $\theta=g_1^{\,h}$. As $\sigma_a g_r=\omega(a)^r g_r$, θ is fixed by H_T, lies in E_T, and has π-valuation h: it is a uniformizer of $E_T\otimes\mathbb{Q}_p$, which is totally ramified of degree e. For the same reason $g_{hj}/\theta^{\,j}$ and $\theta^{\,e}/p$ are fixed by all of Gal, lie in $\mathbb{Q}_p$, and have valuation 0: they are units of $\mathbb{Z}_p$. So $P_{hj}\alpha=u_j\,p^{b_j}\theta^{\,j}=u'_j\,\theta^{\,j+eb_j}$ with $u_j,u'_j\in\mathbb{Z}_p^\times$, and p is a unit times θ^e.

Monomial. R is generated over $\mathbb{Z}_p$ by unit multiples of θ^s, s in the generators of S_T, so $R = \sum_{s\in S_T} \mathbb{Z}_p\theta^s$ inside $\mathcal{O}_{E_T} \otimes \mathbb{Z}_p = \mathbb{Z}_p[\theta]$, whose $\mathbb{Z}_p$-basis is $1, \theta, \dots, \theta^{e-1}$. For each residue i modulo e let $w_i = i + ec_i$ be the least element of S_T in that class (the Apéry set of S_T with respect to e). Then $R = \bigoplus_{i<e} \mathbb{Z}_p\, p^{c_i}\theta^i$ up to units, the valuations of $R \setminus 0$ are exactly S_T, and $v_p[\mathbb{Z}_p[\theta] : R] = \sum_i c_i$, which is the number of gaps of S_T by Selmer's formula [6] .

Evidence & scope

- **Program:** δ(T) as a Hermite-normal-form index equals g(S_T) in every centred spectrum with 3 ≤ n ≤ 6, 5 ≤ p ≤ 31 (2414 orbits), and for n = 7, 8, 9 below the bound (11 331 orbits). `censo.py`

Theorem 4. (the semigroup of moments) p. 4

If $p>(n-1)(n-2)$ (equivalently $p \geq (n-1)(n-2)$, as p is odd), then $S_T = \Gamma_T$ and

$$\setlength\fboxsep6pt\setlength\fboxrule0.6pt \delta(T) = \#\big(\mathbb{N}_0 \setminus \langle e,\ j :\ M_{hj}(T) \neq 0$$

The same holds for every p when $n \leq 12$.

Classical and new: the conductor bound uses Schur's bound on the Frobenius number, as given by Brauer [Bra42].

↳ Used in: Corollary 7 (p. 8), Proposition 12 (p. 10)

Proof

If the conductor $c(\Gamma_T)$ is at most $e+1$, every $j + eb_j$ with $b_j \geq 1$ is $\geq e+1 \geq c$ and already lies in Γ_T. Let $V = \{t^h : t \in T \setminus 0\}$, a set of $d \leq n/h$ elements of the group $\mu_e \subset \mathbb{F}_p^\times$ of h-th powers; each $v \in V$ is hit h times, so $M_{hj}(T) = h\, p_j(V)$. With $A(Y) = \prod_{v\in V}(1 - vY)$ and $B(Y) = \prod_{v\in\mu_e\setminus V}(1 - vY)$, $A(Y)B(Y) = 1 - Y^e$. Newton's identities $p_j + a_1 p_{j-1} + \cdots + ja_j = 0$, with $j<e<p$ invertible, show by induction that the indices $j<e$ with $p_j(V) \neq 0$ and those with $a_j \neq 0$ generate the same semigroup, and likewise for the coefficients of $1/A = B/(1 - Y^e)$, which are those of B below e. So Γ_T is generated by e and a set G of integers at most $D = \min(d, e-d)$: the support of A or that of B, whichever has the smaller degree. The elements of G have no common factor $g>1$: every exponent of A is below e, hence a combination of G alone, so A would be a polynomial in Y^g; its roots, and so V, would be stable under μ_g, and as $T \setminus 0$ is the full preimage of V under $t \mapsto t^h$, T would be stable under $\mu_{gh} \supsetneq H_T$. By Schur's bound on the Frobenius number [4] , $c(\Gamma_T) \leq c(\langle G\rangle) \leq (D-1)(D-2)$, which is 0 if $D \leq 2$; otherwise $3 \leq D \leq d \leq n/h$ and $(D-1)(D-2) \leq (n/h-1)(n/h-2) \leq (n-1)(n-2)/h \leq p/h \leq e+1$ when $p \geq (n-1)(n-2)$.

For $n \leq 12$, exchange A and B if necessary so that $D = \deg A \leq e/2$; then $D \leq n \leq 12$, $G = \operatorname{supp} A \setminus 0$, and $D \in G$, the coefficient of Y^D being $\pm\prod_v v \neq 0$. Moreover $e \in \langle G\rangle$: every monomial of $B = [A^{-1}]_{\leq e-D}$ has its exponent in $\langle G\rangle$, so if $e \notin \langle G\rangle$ the coefficient of Y^e in AB would be 0, not -1. Hence $\Gamma_T = \langle G\rangle$, and since

$e \in \Gamma_T$ it suffices to show $c(\Gamma_T) \leq e$. Let κ be the multiplicity of Γ_T. If $\kappa = 2$ or 3, a generator $u \leq D$ prime to κ gives $c \leq c(\langle \kappa, u \rangle) \leq 2(D-1) < e$. If $\Gamma_T = \langle a, b \rangle$ has two minimal generators, then $ab \leq e$: otherwise every degree up to e has a unique expression $ia + jb$, the substitution $Y^{ia+jb} \mapsto U^i V^j$ turns $AB = 1 - Y^e$ into a factorisation of a binomial $1 - U^r V^s$, all of whose factors are polynomials in one monomial, while A contains both U and V; so $c = (a-1)(b-1) < e$. In general the roots of the reciprocal $F(X) = X^D A(1/X)$ are D elements of μ_e with trivial stabiliser, and for $r \in \mu_e \setminus 1$ the polynomial $F(rX) - r^D F(X)$ is nonzero of degree at most $D - \kappa$; counting the ordered pairs of roots with quotient r gives $D(D-1) \leq (e-1)(D-\kappa)$. Among the semigroups generated by integers at most 12 with multiplicity at least 4 and three or more minimal generators, only $\langle 6, 9, 11 \rangle$, $\langle 6, 10, 11 \rangle$, $\langle 7, 9, 12 \rangle$ and $\langle 8, 11, 12 \rangle$ are compatible with $e \geq 2D$, this inequality and $c(\Gamma_T) > e$, in sixteen triples (D, Γ_T, e). In each of them the truncation $B = [A^{-1}]_{\leq e-D}$ makes a coefficient of AB below degree e equal to a nonzero monomial in the leading coefficients of A — $-2c_6c_9$, $-2c_6c_{10}$, $-c_9^2$, $-2c_{11}c_{12}$ or $3c_8^2c_{11}$ — which contradicts $AB = 1 - Y^e$, since $p > e \geq 23$.

Evidence & scope

- **Program:** the sixteen residual triples (D, Γ_T, e) of the case n ≤ 12, re-enumerated, and their obstructing coefficients computed symbolically. `momentos_D12.py`
- **Program:** S_T = Γ_T in every orbit of the census (2414 + 11 331 orbits). `censo.py`

4 One semigroup, every property

Corollary 5. (the index of a column) p. 5

\setlength\fboxsep6pt\setlength\fboxrule0.6pt *[O_μ:A_μ]=p^ϕ(m_μ) # (N_0 \S_T_μ).*

↳ Not used by another result of the paper.

Corollary

Evidence & scope

- **Program:** checked as an index of integer lattices in 662 columns. `nivel_rango.py`

Corollary 6. (Gorenstein columns) p. 5

A_μ is Gorenstein if and only if S_{T_μ} is symmetric.

NEW **Classical and new:** Kunz's theorem on the value semigroup of a one-dimensional Gorenstein ring [Kun70].

↳ Not used by another result of the paper.

Proof

Away from p the order is maximal. At p, $A_\mu \otimes \mathbb{Z}_p$ is a product over the primes of the unramified $\mathbb{Z}[\zeta_m] \otimes \mathbb{Z}_p$ of copies of $R \otimes W$, W unramified; each is a complete local domain with residue field of the form $\mathbb{F}_{p^f}$, equal to that of its normalisation, and value

semigroup S_T; Kunz's theorem [13 , Thm., p. 749] says such a ring is Gorenstein exactly when its value semigroup is symmetric.

Evidence & scope

- **Program:** the first non-Gorenstein column at prime height, SU(5)_12, μ = (0,0,2,1): [O_μ : A_μ] = 17^2 and [A_μ : f] = 17 as lattice indices, with the control μ = 0. `primera_no_gorenstein.py`

Corollary 7. (the exponent) p. 6

If $c(S_T) \leq e$, then $p\,\mathcal{O}_\mu \subseteq A_\mu$ and $\mathcal{O}_\mu / A_\mu$ is elementary abelian. This holds whenever $p > (n-1)(n-2)$.

↳ Not used by another result of the paper.

Proof

$p\,\mathbb{Z}_p[\theta] = \theta^e \mathbb{Z}_p[\theta]$ lies in the monomial ring R exactly when every integer $\geq e$ is in S_T. For the second sentence, the proof of Theorem 4 gives $S_T = \Gamma_T$ and $c \leq (n-1)(n-2)/h \leq (p-1)/h = e$.

Evidence & scope

- **Program:** c(S_T) ≤ e and p·O_E_T ⊆ C_T on the lattice, in the 2414 orbits of the census and in the 2644 with 3 ≤ n ≤ 8, p ≤ 23. `exponente.py`

Corollary 8. (the vacuum) p. 6

The column $\mu = 0$ is maximal: $A_0 = \mathcal{O}_0$ for every $\mathrm{SU}(n)_k$ with $n + k$ prime.

Classical and new: agrees with the preceding note [I], where at prime height the column μ = 0 lies outside every family of non-maximal segments.

↳ Not used by another result of the paper.

Proof

For $\mu = 0$, $\ell_i = n - i$ and T is a centred arithmetic progression, $-T = T$, so $h \geq 2$; for $k \geq 2$ its second moment is $M_2 = n(n^2-1)/12 \neq 0$ in $\mathbb{F}_p$, since $2 \leq n \leq p-2$. Then $1 \in \Gamma_T$ and $S_T = \mathbb{N}_0$; here $h = 2$ exactly, since $M_2 \neq 0$ and $M_r = 0$ unless $h \mid r$. For $k = 1$, T is $\mathbb{F}_p$ minus a point, $C_T = \mathbb{Z}$.

Proposition 9. (level–rank) p. 6

Under level–rank duality $\mathrm{SU}(n)_k \leftrightarrow \mathrm{SU}(k)_n$, the spectrum of the transposed column is $T' = -(\mathbb{F}_p \setminus T)$, and $C_{T'} = \sigma_{-1}(C_T)$: the semigroup, the index δ, the conductor and the Gorenstein property are preserved. Only the central factor $\varphi(m)$ may change.

↳ Not used by another result of the paper.

Proof

For a partition in an $n \times k$ box, the sets $\{\lambda_i + n - i\}$ and $\{n - 1 + j - \lambda'_j\}$ are complementary in $\{0, \ldots, n+k-1\}$ [5 , I.(1.7)] ; reducing modulo $p = n + k$ and centring gives $T' = -(\mathbb{F}_p \setminus T)$. Next, $\prod_{t \in T}(1 - y_t X) \cdot \prod_{t \notin T}(1 - y_t X) = 1 - X^p$, so the elementary symmetric functions of $y_{\mathbb{F}_p \setminus T}$ are the coefficients of a power series with integral coefficients in those of y_T, and conversely: $C_{\mathbb{F}_p \setminus T} = C_T$. Finally σ_{-1}, complex

conjugation, carries $C_{\mathbb{F}_p\setminus T}$ to $C_{T'}$, and $\sigma_{-1}\theta = \omega(-1)^h\theta = \pm\theta$, so it preserves the monomial structure and S_T.

Evidence & scope

- **Program:** three level-rank pairs, every column, classes (h, gaps) in the proportion n/k. `nivel_rango.py`

Proposition 10. (the conformal weight) p. 6

Let $\Delta_\mu = \big(|\mu+\rho|^2 - |\rho|^2\big)/2p$ be the conformal weight of μ, with roots of square length 2, so that $|\rho|^2 = n(n^2-1)/12$ and $e^{2\pi i\Delta_\mu}$ is the eigenvalue of the modular matrix $\mathcal{T}$ at μ up to the central charge [11 , Thm. 13.8] . Then

$$\setlength\fboxsep6pt\setlength\fboxrule0.6pt\ M_2(T_\mu) \equiv |\mu+\rho|^2 \pmod{p},$$

so $M_2(T_\mu) = 0$ if and only if p does not divide the order of the root of unity $e^{2\pi i\Delta_\mu}e^{\pi i|\rho|^2/p} = e^{\pi i|\mu+\rho|^2/p}$. If moreover H_T is trivial, then $\delta(T) = 1$ if and only if $M_2(T)M_3(T) \neq 0$; so the index of the column is $p^{\varphi(m)}$ exactly when both moments are nonzero, and the conformal weight decides the first of the two.

↳ Not used by another result of the paper.

Proof

In the coordinates of §2, $\mu+\rho$ is $(\ell_1,\dots,\ell_n)$ modulo $(1,\dots,1)$, so $|\mu+\rho|^2 = \sum_i(\ell_i - L/n)^2 = \sum_i \ell_i^2 - L^2/n$. With $t_i = \ell_i - vL$ and $vn \equiv 1$, $\sum_i t_i^2 \equiv \sum_i \ell_i^2 - 2vL^2 + nv^2L^2 \equiv \sum_i \ell_i^2 - L^2/n$. The root of unity is $e^{2\pi iQ/2np}$ with $Q = n|\mu+\rho|^2 \in \mathbb{Z}$, and as $p \nmid 2n$ its order is prime to p exactly when $p \mid Q$. For the second sentence, $h = 1$ gives $e = p-1 \geq 4$. The only semigroup of genus one is $\langle 2,3\rangle$. By Theorem 3 an integer below e lies in S_T exactly when it is a sum of indices $j<e$ with $M_j \neq 0$; $M_1 = 0$, so $1 \notin S_T$, and 2 and 3 lie in S_T exactly when M_2 and M_3 are nonzero, and then $S_T \supseteq \langle 2,3\rangle$ has at most one gap.

Evidence & scope

- **Program:** the congruence $M_2 \equiv |\mu+\rho|^2$ in all 1820, 4368 and 1540 columns of SU(5)_12, SU(6)_11, SU(4)_19; the genus-one criterion in every orbit with $p \leq 31$, $n \leq 9$. `peso_conforme.py`

5 How many columns of each kind

Proposition 11. (the maximal columns) p. 7

Let $2 \leq n \leq p-2$, and for $h \mid p-1$ let $e = (p-1)/h$, $G_e \subset \mathbb{F}_p^\times$ the subgroup of order e, and $Z_{p,e}(d)$ the number of d-subsets of G_e with sum 0. The number of columns of $\mathrm{SU}(n)_{p-n}$ with $S_T = \mathbb{N}_0$ is

$$\setlength\fboxsep{6pt}\setlength\fboxrule{0.6pt}\ N_0(p,n) = n \sum_{\substack{h\mid p-1,\ h>1 \\ n \bmod h \in \{0,1\}}} \left[\binom{(p-1)}{\lfloor n/h \right.$$

↳ Not used by another result of the paper.

Proof

> $S_T = \mathbb{N}_0$ exactly when $1 \in S_T$, and a generator $j + eb_j$ equals 1 only for $j = 1$, $b_1 = 0$, that is when $M_h(T) \neq 0$. If $h = 1$ this fails, $M_1 = \sum T = 0$. For $h>1$, $T \setminus 0$ is a union of d cosets of H_T, so $n = hd$ or $hd + 1$ according as $0 \notin T$ or $0 \in T$, and every such set is centred; and $M_h = h \sum_{u \in U} u$ with $U = \{t^h\} \subset G_e$ of size d. If $\sum U \neq 0$ the stabiliser is exactly of order h, since a larger one would kill M_h. So the sets with $S_T = \mathbb{N}_0$ and stabiliser of order h are counted by $\binom{e}{d} - Z_{p,e}(d)$, the families for different h are disjoint, and each centred spectrum stands for n columns.

Evidence & scope

- **Program:** the formula against enumeration: 30 of 30 pairs (p, n), p = 17, 19. `maximales.py`

6 The first column that is not Gorenstein

Proposition 12. (the largest defect) p. 8

> *For every centred T of size $n \geq 3$, $\delta(T) \leq (n-1)(n-2)/2$, with equality exactly when $\prod_{t \in T}(X - t) = X^n + aX + b$ with $ab \neq 0$; then $\Gamma_T = S_T = \langle n-1, n \rangle$. Such T exist for every sufficiently large prime p.*

✅ Checked when this page was built: GenusBound.lean compiles with Lean 4.35.0, no errors, no sorry. Axioms — genus_le_D, eq_pair_of_genus_eq, genus_pair_n: propext, Classical.choice, Quot.sound. Reproduce: Lean 4.35.0, Mathlib f0b4c8150aa0; in that Lean project, `lake env lean GenusBound.lean` (SHA-256 of the compiled file: `3c315a332e4ffcc2…`; a source page on this site may show it with comments edited, under its own hash).

> NEW **Classical and new:** existence for large p from the count of Bank, Bary-Soroker and Rosenzweig [BBR15, Thm. 2.3].

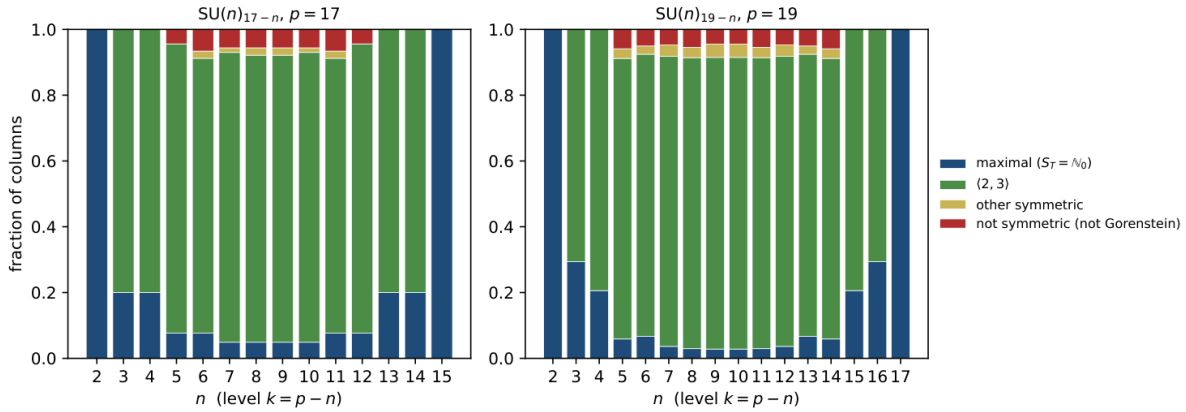
↳ Not used by another result of the paper.

Proof

> A numerical semigroup of multiplicity κ generated in degrees at most D has its Apéry elements modulo κ at most $D, 2D, \ldots, (\kappa-1)D$, so its genus is at most $(\kappa-1)(D-1)/2 \leq (D-1)(D-2)/2$; with $\delta(T) = g(S_T) \leq g(\Gamma_T)$ and $D \leq n$ this is the bound. Equality forces $D = n$, $h = 1$, $0 \notin T$ and $\kappa = n - 1$, hence $\Gamma_T = \langle n-1, n \rangle$, and by Newton the only coefficients of $\prod(X - t)$ besides the leading one sit in degrees 1 and 0. Conversely, for a trinomial with n distinct roots the quotients of distinct roots are all different (if $x = ry$ with $r \neq 1$, then $a(r - r^n)y + b(1 - r^n) = 0$ fixes y), so $n(n-1) \leq p - 2$, the conductor

> $(n-1)(n-2)$ is below $e=p-1$, and $S_T=\Gamma_T$. Existence: Bank, Bary-Soroker and Rosenzweig [3 , Thm. 2.3] count the polynomials of a given factorisation type in the short interval $f+\mathbb{F}_p[X]_{\le m}$; we use it with $f=X^n$, $m=1$, which they allow when $p\nmid n(n-1)$, as here since $n<p$, and with the type $(1,\dots,1)$. The number of pairs (a,b) for which X^n+aX+b splits into linear factors is then $p^2/n!+O_n(p^{3/2})$, while $ab=0$ or a repeated root excludes $O(p)$ of them.

Figure 2



The columns of $\mathrm{SU}(n)_{p-n}$ by the semigroup S_T of their spectrum, computed with the exact formula of Theorem 3; each centred spectrum stands for n columns. Maximal columns (blue) have $A_\mu=\mathcal{O}_\mu$; $\langle 2,3\rangle$ (green) has one gap; the other symmetric semigroups (gold) are Gorenstein with more gaps; the red ones are not Gorenstein. The bars for n and $p-n$ agree, as level–rank duality demands.

Evidence & scope

- **Program:** n = 4, p ≤ 71: every split trinomial gives δ = 3 and conversely; n = 5, p ≤ 71: only p = 67, all 66 give δ = 6. `maximo.py`

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